

11/3/23

TASK 2-1

DDL X DML Commands.

Aim:- To perform (DDL) data definition language and (DML) Data manipulation language commands.

DDL Commands:-

- Create
- Drop
- Alter
- Truncate
- Rename

DML Commands:-

- Insert
- Update
- Delete
- SELECT

DDL Commands:-

- * Creating a Table by using Create Commands.

```
CREATE TABLE student (
```

```
stu-id int,
```

```
stu-name varchar(30),
```

```
stu-department varchar(30),
```

```
stu-gender varchar(30),
```

```
stu-ph-no int;
```

- * Using ALTER Command we can add or

Remove the Column.

SYNTAX:-

```
ALTER table student ADD Column stu-depart-id  
int;
```

After Alter table

desc student

STU - ID	INT
STU - NAME	VARCHAR(30)
STU - DEPARTMENT	VARCHAR(30)
STU - GENDER	VARCHAR(15)
STU - PH - NO	INT
STU - DEPART - D	INT

After inserting into table.

select * from student:-

S.NO.	STU - ID	STU - NAME	STU - DEPART	STU - GENDER
1.	30628	Shravan	CSE	Male
2.	28800	Mirille	CSE	Male.

STU - PH - NO	STU - DEPART - ID
8688056	1225
123456	1425

* DROP

DROP command is used to drop the table completely.

DROP TABLE student.

TRUNCATE

* Truncate command is used to remove all data but keep structure.

TRUNCATE TABLE student.

DML Commands

INSERT

* Insert Command is used to insert the values to the table.

INSERT INTO student VALUES

(30628, 'Sriram', 'CSE', 'Male', 9682426)

(28800, 'Gopal', 'CSE', 'Male', 8624321)

UPDATE

* Update Command is used to update the existing records

UPDATE student

SET stu_name = 'Sriram'

WHERE stu_ID = 28800;

DELETE:-

* Delete Command is used to delete a record.

DELETE FROM student

WHERE stu_ID = 28800;

After updating the table
select * from student.

S.No.	stu. id	stu-name Shravan	stu-department	stu-gender
1.	30628	Shravan	CSE	Male
2.	28800	Mohith	CSE	Male

stu-phone-no	stu-depart-id
868056	1225
123456	1425



SELECT

* Select command is used to retrieve the records from table.

SELECT * FROM student

S.NO	STU-ID	STU-name	STU-department	STU-Gender	STU-Ph-no
1.	30628	Srixam	CSE	Male	928628

Ex:-

CREATE TABLE students (

Roll-no INT,

Name ~~VAR~~ VARCHAR(30),

Age INT,

COURSE VARCHAR(30));

ALTER TABLE students ADD

Email VARCHAR(50);

INSERT INTO students VALUES

(1, 'Srixam', 19, 'BTech', 'srixam@gmail.com')

(2, 'Roy', 21, 'BTech', 'roy@gmail.com')

(3, 'Arjun', 20, 'BTech', 'arun@gmail.com')

UPDATE students

SET Email = Vtu 30628@gmail.com

WHERE Roll-no = 1;

DELETE FROM students

WHERE Roll-no = 2;

SELECT * FROM students;

S.NO	Roll-no	Name	Age	Course	Email
1	1	Sixam	19	B.TECH	Sixam@gmail.com
2	3	Roy	21	B Tech	roy@gmail.com

SELECT Name FROM students;

S.NO	Name
1	Sixam
2	Roy

SELECT * FROM students
WHERE Name = 'Sixam';

S.NO	Roll-no	Name	Age	Course	Email
1	1	Sixam	19	B.Tech	Sixam@gmail.com.

VEL TECH	
EX No.	2.1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	4
RECORD (5)	1
TOTAL (20)	16
DATE	

Result:- All the DDL and DML Commands are
SQL are successfully executed.

11/9/23 TASK (2.2):- DDL & DML commands with constraints

Aim:- To perform DDL and DML command with constraints in SQL.

CONSTRAINTS:-

- NOT NULL
- UNIQUE
- PRIMARY KEY
- FOREIGN KEY
- CHECK
- DEFAULT

NOT NULL

It ensures a column cannot store NULL values.

SYNTAX

```
CREATE TABLE TABLE-NAME(  
    COLUMNNAME DATATYPE NOTNULL);
```

UNIQUE

It ensures all values in a column are unique.

SYNTAX

```
CREATE TABLE table name (  
    column name datatype UNIQUE);
```

PRIMARY KEY

It is the combination of NOT NULL & UNIQUE

FOREIGN KEY

It ensures values in one table match values in another table.

SYNTAX

FOREIGN KEY (columnname) REFERENCES another table name
(column name).

DEFAULT:-

It provides a default value for a column when none is specified.

Example:-

```
CREATE TABLE DEPARTMENT
```

```
DEPT_ID INT PRIMARY KEY,
```

```
DEPT_NAME VARCHAR(20) UNIQUE NOT NULL);
```

```
CREATE TABLE STUDENT
```

```
STU_ID INT PRIMARY KEY,
```

```
STU_NAME VARCHAR(30) NOT NULL;
```

```
STU_DEPARTMENT INT DEFAULT 101,
```

```
STU_GENDER VARCHAR(6)
```

```
CHECK (STU_GENDER IN ('MALE', 'FEMALE'))
```

```
STU_PH_NO BIGINT UNIQUE,
```

```
FOREIGN KEY (STU_DEPARTMENT) REFERENCES
```

```
DEPARTMENT (DEPT_ID);
```

```
INSERT INTO DEPARTMENT VALUES
```

```
(101, 'CSE');
```

```
(102, 'ECE');
```

```
(103, 'IT');
```

```
INSERT INTO STUDENT VALUES
```

```
(1, 'RAVI', 102, 'MALE', 9876543210),
```

```
(2, 'ANITA', 'FEMALE', 9876501234);
```

SELECT * FROM STUDENT - Before performing ALTER Command.

S.NO	STU-ID	STU-NAME	STU-DEPARTMENT	STU-GENDER	STU-PH-NO.
1	1	RAVI	102	MALE	9876543210
2	2	ANITA	101	FEMALE	9876501234

SELECT * FROM DEPARTMENT - Before performing ALTER Command.

S.NO	STU-ID	
1	101	CSE
2	102	ECE
3	103	IT

SELECT * FROM STUDENT - after performing update command.

S.NO	STU-ID	STU-NAME	STU-DEPARTMENT	STU-GENDER	STU-PHNO	STU-EMAIL
1	1	RAVI	102	MALE	9876543210	NULL
2	2	ANITA	103	FEMALE	9876501234	NULL

SELECT * FROM STUDENT - after performing delete command.

S.NO	STU-ID	STU-NAME	STU-DEPARTMENT	STU-GENDER	STU-PH-NO	STU-EMAIL
1	1	RAVI	102	MALE	9876543210	NULL

SELECT * FROM DEPARTMENT;

SELECT * FROM STUDENT;

ALTER TABLE STUDENT

ADD STU-EMAIL VARCHAR(50) DEFAULT

'vishal@gmail.com';

S.NO	STU-ID	STU-NAME	STU-DEPARTMENT	STU-GENDER	STU-PH-NO	STU-EMAIL
1	1	RAVI	102	MALE	9876543210	NULL
2	2	ANITA	101	FEMALE	9876501234	NULL

UPDATE STUDENT

SET STU-DEPARTMENT = 103

WHERE STU-NAME = 'ANITA'

DELETE FROM DEPARTMENT

WHERE DEPT-ID = 103,

INSERT INTO STUDENT VALUES

(2, 'ANITA', 'FEMALE', 9876501234);

(7, 'Shreshtha', 'FEMALE', 9876501230);

DROP TABLE DEPARTMENT;

----- Error -----

Could not drop object 'DEPARTMENT' because it is referenced

By foreign key constraint.

To solve this first we have to drop student table after
we have to drop department table.

SELECT * FROM STUDENTS -- AFTER inserting values.

S.NO	STU-NAME	STU-DEPARTMENT	STU-GENDER	STU-PH-NO	STU-EMAIL
1.	RAVI	102	MALE	9876543210	NULL
2.	ANITA	101	FEMALE	9876501234	veltech@gmail.com
3.	Shreshtha	101	MALE	9876501230	veltech@gmail.com

SELECT * FROM DEPARTMENT: - After inserting values.

S.NO	DEPT-ID	DEPT-NAME
1	101	CSE
2	102	ECE
3	103	IT

VELTECH	
EX No.	2-2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	4
RECORD (5)	2
TOTAL (20)	14
SIGN WITH DATE	21/8/23

RESULT:- All the DDL and DML commands are with constraints are performed and executed successfully.